PRAVEEN KUMAR MURALI

Robotics MSc Student | ROS 2, Perception, SLAM, Planning and Control

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PROFILE

Erasmus Mundus master's student in Intelligent Field Robotic Systems, specialising in perception, SLAM, planning and control for mobile robots in ROS 2. Before the master's, worked for three years as a robotics engineer, deploying AGVs and autonomous mobile robots in factories and warehouses with ROS, ROS 2 and Nav2. Focused on building robots that work reliably and safely outside of simulation.

EDUCATION

Erasmus Mundus Joint Master in Intelligent Field Robotic Systems (IFRoS) Current
ELTE, Budapest, Hungary (3rd semester) | University of Girona, Spain (1st year)

Relevant courses: Probabilistic Robotics, Autonomous Systems, Hands-on Planning, Hands-on Localization, Hands-on Intervention, Deep Network Developments, Machine Learning

B.E. Mechatronics Engineering Aug 2019 – May 2023
Dr. Mahalingam College of Engineering and Technology, Coimbatore, India | GPA 8.9/10

PUBLICATIONS

M. Yamin, **P. K. Murali**, A. Das, H. Rajani, A. Saha, et al. *Automated Skin Lesion Detection in Total Body Photography: A Multi-Architecture Benchmark*. ISIC Workshop, MICCAI 2026 (spotlight paper).
papers.miccai.org/miccai-2026-sat/paper/ISIC_016.pdf

ROBOTICS PROJECTS (MSc)

Autonomous Navigation Stack on a TurtleBot 4 Mar – May 2026
Hands-on Planning project | ROS 2 Jazzy, Python, RTAB-Map, LiDAR | team of three

- Built an RRT* global planner and a real-time DWA local planner (10 Hz), both from scratch (no Nav2), with frontier-based exploration on top of RTAB-Map LiDAR SLAM.
- Every path is checked against an inflated map and the live LiDAR scan, and the robot recovers on its own when it gets stuck.
- The stack ran across the robot and a laptop over Wi-Fi, so we had to sort out clock drift between the two machines (chrony), a ROS 2 QoS mismatch and wrong scan timestamps.
- In the course lab version, the real robot replanned within about a second when a person walked into its path.

Pick, Transport and Place with a Mobile Manipulator Mar – May 2026
Hands-on Intervention project | ROS 2, Python, OpenCV, TurtleBot 2 + uArm | team of three

- 13-state state machine that finds a box with ArUco markers, picks it up with a suction gripper, carries it to a drop-off point and puts it down.
- Base and arm move together under task-priority control. Joint limits have top priority, so no other task can break them. Tested in simulation and on the real robot.

Graph SLAM for Indoor Localization Mar – May 2026
Hands-on Localization project | ROS 2, GTSAM (iSAM2), Python | team of three, my part: graph optimisation, validation, real-robot tests

- Factor-graph SLAM from 2D LiDAR wall lines (RANSAC), with a custom GTSAM factor and IMU heading.
- Landmark matching with a 95% chi-square test. Ran a controlled experiment with and without data association on the exact same scripted path.

WORK EXPERIENCE

Research Intern | ViCOROB (Computer Vision and Robotics Institute), University of Girona Jun 2026 – Sep 2026
Supervisors: Dr. Anup Saha, Dr. Rafael Garcia

- Built a graph neural network for melanoma risk where each patient is one node and similar patients are linked, using clinical and lesion data from 472 patients.
- Evaluated performance with cross-validation, ablation studies, a simpler baseline model and bootstrap 95% confidence intervals (1,000 resamples).

Engineer, later Centre In-Charge | Forge Innovation and Ventures, India Jun 2023 – Jul 2025

- Led the technical work on 20+ robotics and vision projects for startups and industry partners, cutting their time to market by up to 30%.
- Automation for Tier-1 clients (Ashok Leyland, Elkayem, Microtech), raising production-line efficiency by about 25%.
- Trained 2,000+ students and engineers in robotics and ran hackathons with 500+ participants.

Technical Intern | Forge Innovation and Ventures, Hosur, India Feb 2023 – May 2023

- Deployed a ROS-based AGV in small factories (10 kg payload, 0.5 m/s), cutting manual handling time by 40%.
- Integrated ROS 2 and the Nav2 stack with Maxbyte Technologies, which made later AGV deployments 20% faster.

Robotics Intern | Roboto AI Technologies, Coimbatore, India Aug 2022 – Jan 2023

- 2D LiDAR sensor fusion for an 80 kg warehouse robot (1 m/s), with collisions close to zero.

OTHER PROJECTS

- **Indoor and outdoor robot (2024 – 2025):** Ackermann robot with LiDAR, IMU and encoder fusion, SLAM indoors and RTK-GPS outdoors.
- **Retrofit autonomy module (2024 – 2025):** started a kit that lets existing electric tow motors drive on their own, covering sensors, control and navigation.
- **Wildlife monitoring with edge AI (2023 – 2024):** YOLOv8 on the device, running 24/7 for an NGO.
- **5-axis SCARA robot (2023 – 2024):** 80% 3D-printed, ±0.02 mm repeatability, own kinematics and firmware.
- **Milk quality predictor:** Random Forest model built into commercial vending machines to check milk quality in real time.

SKILLS

Programming: Python, C++, Git, Linux

Robotics: ROS 2, Nav2, RViz, Gazebo, Stonefish, TurtleBot 4 and 2, RTAB-Map, GTSAM, LiDAR, cameras

Autonomy stack: SLAM, EKF, multi-sensor fusion (LiDAR, IMU, wheel encoders), RRT, RRT*, A*, DWA, frontier exploration, occupancy grids, task-priority control, state machines

Networked robots: multi-machine ROS 2 over Wi-Fi, QoS profiles, clock sync (chrony)

Probability and statistics: Bayes and Kalman filters, particle filters, Monte Carlo sampling, uncertainty propagation, chi-square and Mahalanobis tests, bootstrap confidence intervals, cross-validation

Machine learning: PyTorch, PyTorch Geometric (graph neural networks), scikit-learn, YOLO, OpenCV, Random Forest, reinforcement learning (Q-learning, DQN, policy gradients)

Industrial robots: UR10e (URScript), Stäubli TX60 and TS60 (VAL3), KUKA, FANUC

Embedded and edge: Jetson Nano, Raspberry Pi, ESP32, Arduino, on-device inference

Languages: English (fluent), Tamil (native), Kannada (fluent), Hindi (intermediate), Spanish (basic)